

Program 6:

Deploy an Automated Build Pipeline using Docker Hub

Step 1: Create Project Directory

```
mkdir program6
cd program6
```

Step 2: Create Node.js Application Files

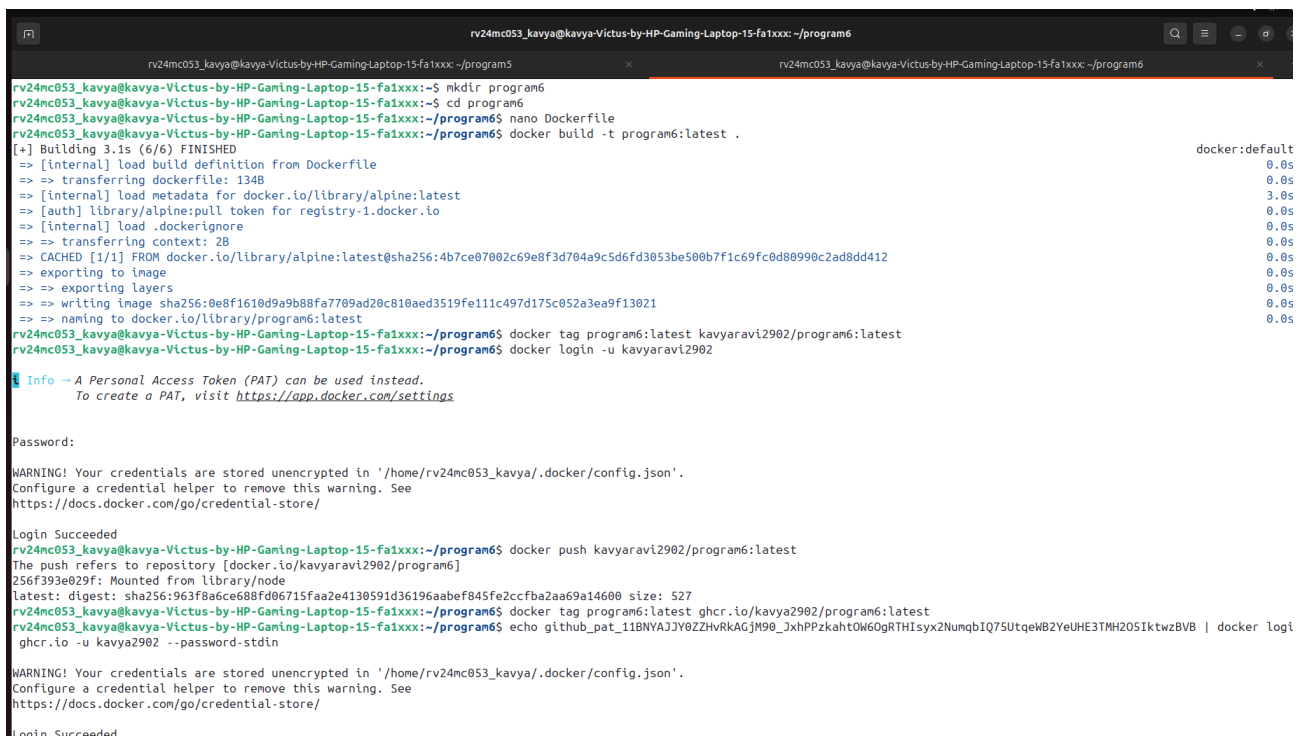
Create package.json with a start script and dependencies.

Create app.js containing a simple Express server.

Step 3: Create Dockerfile

Step 4: Build Docker Image

```
docker build -t program6 .
```



```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program6
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program6$ mkdir program6
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program6$ cd program6
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program6$ nano Dockerfile
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program6$ docker build -t program6:latest .
[+] Building 3.1s (6/6) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 134B
=> [internal] load metadata for docker.io/library/alpine:latest
=> [auth] library/alpine:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> CACHED [1/1] FROM docker.io/library/alpine:latest@sha256:4b7ce07002c69e8f3d704a9c5d6fd3053be500b7f1c69fc0d80990c2ad8dd412
=> exporting to image
=> => writing image sha256:0e8f1610d9a9b88fa7709ad20c810aed3519fe11c497d175c052a3ea9f13021
=> => naming to docker.io/library/program6:latest
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program6$ docker tag program6:latest kavyaravi2902/program6:latest
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program6$ docker login -u kavyaravi2902

Info -> A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings

Password:

WARNING! Your credentials are stored unencrypted in '/home/rv24mc053_kavya/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program6$ docker push kavyaravi2902/program6:latest
The push refers to repository [docker.io/kavyaravi2902/program6]
256f393e029f: Mounted from library/node
latest: digest: sha256:963f8a6ce688fd06715faa2e4130591d36196aabef845fe2ccfba2aa69a14600 size: 527
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program6$ docker tag program6:latest ghcr.io/kavya2902/program6:latest
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program6$ echo 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 | docker login
ghcr.io -u kavya2902 --password-stdin

WARNING! Your credentials are stored unencrypted in '/home/rv24mc053_kavya/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
```

Step 5: Run Docker Container Locally

```
docker run program6
```

Step 6: Login to Docker Hub Using Access Token

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program6$ echo ghp_cXMnStb4iISvF1ge3cnJjuiqFxh4XI2mJ9Tz | docker login ghcr.io -u kavya2902 --password-stdin
WARNING! Your credentials are stored unencrypted in '/home/rv24mc053_kavya/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program6$ docker push ghcr.io/kavya2902/program6:latest
The push refers to repository [ghcr.io/kavya2902/program6]
256f393e029f: Pushing [=====] 6.491MB/8.322MB

user
256f393e029f: Pushed
latest: digest: sha256:963f8a6ce688fd06715faa2e4130591d36196aabef845fe2ccfba2aa69a14600 size: 527
```

Step 7: Tag Docker Image for Docker Hub

Step 8: Push Docker Image to Docker Hub

Step 9: Initialize Git Repository and rename the branch as main

git init

git branch -m main

```
Initialized empty Git repository in /home/rv24mc053_kavya/program6/.git/
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program6$ git add .
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program6$ git commit -m "Initial commit for program6"
[master (root-commit) 05c3ee1] Initial commit for program6
1 file changed, 5 insertions(+)
create mode 100644 Dockerfile
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program6$ git remote add origin https://github.com/kavya2902/program6.git
```

Step 10: Add GitHub Remote Repository

Step 11: Stage Project Files

git add .

Step 12: Commit Source Code

git commit -m "Program 6 - Automated Build Pipeline"

Step 13: Push Source Code to GitHub

git push -u origin main

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program6$ git push -u origin main
Username for 'https://github.com': kavya2902
Password for 'https://kavya2902@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 319 bytes | 319.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/kavya2902/program6.git
 * [new branch] main -> main
branch 'main' set up to track 'origin/main'.
```

Step 14: Check if the repository is on git

